

Empirical Mode Decomposition, Ensemble EMD, and CEEMDAN:

A Nonlinear Multiresolution Analysis with Stochastic Regularization

Prepared for an audience of mathematical physicists and analysts working in functional and numerical analysis with a focus on nonlinear methods.

1. Introduction

The decomposition of a time series into oscillatory components is, at bottom, a problem in multiresolution analysis. The classical answers — the Fourier transform, the short-time Fourier transform, and wavelet expansions — all rest on a *fixed, linear, and shift-invariant* basis: a family of global sinusoids, or of sinusoids modulated by a fixed window, or of dilates and translates of a single mother function. These bases are optimal for signals whose oscillatory content is harmonic, stationary, or self-similar across scales, respectively. They are not optimal for the broad class of signals that arise in fluid turbulence, geophysics, cardiac and neural dynamics, or the output of nonlinear dynamical systems, where the instantaneous amplitude and frequency vary in a way that is neither periodic nor self-similar and that no single fixed basis captures without a loss of local information.

Empirical Mode Decomposition (EMD) was introduced by Huang, Shen, Long, Wu, Shih, Zheng, Yen, Tung, and Liu in 1998 [1] precisely to address this gap. Its defining property is that it prescribes **no basis at all**: the oscillatory components — the *intrinsic mode functions* (IMFs) — are extracted adaptively from the data by a purely local, nonlinear sifting procedure. The motivation is the Hilbert-Huang transform (HHT): given an IMF, the Hilbert transform yields a well-defined instantaneous frequency, so a decomposition into “well-behaved” oscillations is the prerequisite for a meaningful time-frequency representation of a nonlinear, nonstationary signal.

The price of this data-adaptivity is that the method is a **nonlinear operator** on the signal space, and nonlinear operators do not come with the convergence, stability, and uniqueness guarantees that linear spectral methods enjoy. As originally formulated, EMD has no general existence or convergence theorem, its stopping rule is ad hoc, and it exhibits well-known pathologies — most notably *mode mixing* — that limit its reliability. A family of refinements addresses these pathologies by a *stochastic regularization* of the nonlinear operator: one perturbs the signal with white noise, decomposes each perturbed copy, and averages. This essay develops that family — EMD, **Ensemble EMD (EEMD)** [2], **Complementary EEMD (CEEMD)** [3], **Complete EEMD with Adaptive Noise (CEEMDAN)** [4], and **Improved CEEMDAN (ICEEMDAN)** [5] — from a functional- and numerical-analysis standpoint, separating carefully what is rigorously established (the convergence of the *linear* iterative-filtering idealization of sifting, and the Monte-Carlo error scaling of the ensemble methods) from what remains heuristic (the convergence of the fully nonlinear spline sifting, the optimality of the stopping rule). It closes with a comparison to the modern competitors **Variational Mode Decomposition (VMD)** [6] and **synchrosqueezing** [7].

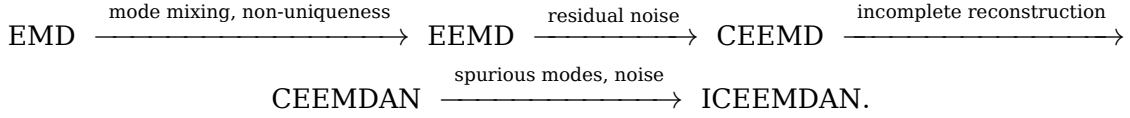
A single **running example** is carried through the whole essay: a two-tone amplitude- and frequency-modulated signal

$$x(t) = a_1(t) \cos(\varphi_1(t)) + a_2(t) \cos(\varphi_2(t)), \quad (1)$$

where the two instantaneous frequencies $\varphi'_1(t)$ and $\varphi'_2(t)$ are *close* over a sub-interval $I_0 \subset [0, T]$ (a frequency-overlap region) and *separated* elsewhere, and where a_1 and a_2 vary slowly. This is the minimal signal on which the distinction between the methods becomes visible: it is simple enough to reason about, and rich enough to exhibit mode mixing, which is the central pathology that the ensemble methods are designed to cure.

1.1 The family tree

The methods form a lineage, each step motivated by a specific failure of its predecessor:



- **EMD** [1] is the raw nonlinear decomposition: adaptive, basis-free, but mode-mixed and non-unique.
- **EEMD** [2] is a *stochastic regularization*: average the EMD over an ensemble of white-noise-perturbed copies. Cures mode mixing at a Monte-Carlo rate $1/\sqrt{N}$, but leaves residual noise.
- **CEEMD** [3] adds the noise in *opposite-sign pairs* $(+n_i, -n_i)$ so that the noise cancels in the ensemble mean, reducing the residual-noise floor.
- **CEEMDAN** [4] constructs the IMFs *sequentially* from ensemble averages with a single, shared residual, enforcing **exact reconstruction** $x = \sum_k c_k + R$, and injects *adaptive* (octave-matched) noise at each stage.
- **ICEEMDAN** [5] refines CEEMDAN by scaling the injected noise to the *standard deviation of the first-mode ensemble*, reducing spurious modes and residual noise further.

The progression is driven almost entirely by two goals: reducing **mode mixing** and improving **reconstruction consistency**. Section 9 situates the family relative to the variational (VMD) and reassignment (synchrosqueezing) approaches.

2. The EMD algorithm

2.1 The intrinsic mode function

An **intrinsic mode function** (IMF) is a function $h(t)$ satisfying two conditions [1]:

1. (*counting*) the number of local extrema and the number of zero crossings in the whole record are equal, or differ by at most one;
2. (*local zero mean*) the mean of the values defined by the upper envelope (the spline through the local maxima) and the lower envelope (the spline through the local minima) is zero, i.e. the two envelopes are symmetric about the local mean.

Condition (1) is a *local* AM-FM regularity requirement: it forces h to be a “single” oscillation at every local time scale, so that an instantaneous frequency $\omega_h(t) = \varphi'_h(t)$ defined by the zero-crossing rate is meaningful. Condition (2) is the *zero-mean* requirement that removes the DC offset and, to first order, the slow amplitude trend. In the language of the HHT, an IMF is a function of the form $a(t) \cos \varphi(t)$ whose amplitude a and phase φ are slowly varying relative to the oscillation [1,10].

A mathematical caveat is in order. The IMF conditions are a *heuristic* guarantee that the Hilbert transform yields a meaningful instantaneous frequency; they are not a theorem. Rigorous instantaneous-frequency theory for a product $a(t) \cos \varphi(t)$ requires **Bedrosian-type** separation assumptions — the amplitude a low-pass and the phase φ high-pass, with the two spectral supports disjoint (equivalently, the amplitude varies slowly enough relative to the carrier) — so that the Hilbert transform of the product splits into the product of the Hilbert transforms. The IMF conditions are an empirical proxy for this; mono-component behavior remains partly heuristic, and a mathematical physicist should read the HHT's “well-defined instantaneous frequency” as a *working assumption*, not a consequence of the two IMF conditions alone.

Note that the IMF is **not defined by a linear constraint** (it is not the range of a projection, and it is not a member of a fixed subspace). It is defined by a *data-dependent* combinatorial and local-symmetry condition. This is the first point at which EMD departs from every linear spectral method, and it is the source of both its power and its analytical difficulty.

2.2 The sifting process

Given a signal $x(t)$, EMD extracts the *fastest* oscillation by an iterative **sifting** process [1]. Starting from the current residual $r^{(0)} = x$:

1. **Locate** all local maxima and minima of $r^{(k)}$.
2. **Interpolate** the upper envelope $E_+(r^{(k)})$ by a spline through the maxima, and the lower envelope $E_-(r^{(k)})$ by a spline through the minima.
3. **Subtract the mean envelope**:

$$r^{(k+1)} = r^{(k)} - \frac{1}{2} \left(E_+(r^{(k)}) + E_-(r^{(k)}) \right). \quad (2)$$

4. **Repeat** steps 1–3 until a stopping criterion (Section 2.3) is met; the output of the sifting is the first IMF, C_1 .
5. **Subtract** C_1 from x to form the first residual $R_1 = x - C_1$, and **recurse**: apply the same sifting to R_1 to obtain C_2 , and so on.

The process terminates when the residual R_K is monotone (or has fewer than two extrema), from which no further IMF can be extracted. The result is the decomposition

$$x(t) = \sum_{k=1}^K C_k(t) + R_K(t), \quad (3)$$

with C_1 the fastest mode, C_K the slowest, and R_K the trend.

In **running example (1)**, the first sifting step is meant to isolate the fastest of the two tones. Where the two instantaneous frequencies are well separated, this works as intended: C_1 tracks the faster tone φ_1 and C_2 tracks the slower tone φ_2 . The difficulty is the overlap region I_0 .

2.3 Stopping criteria

The sifting (2) is an *infinite* iteration in principle; a stopping rule is required. Two criteria are standard.

Cauchy / standard-deviation (SD) criterion [1]. Let $h^{(k)}$ denote the k -th sifting iterate (so $h^{(0)} = r^{(0)}$ and $h^{(k+1)}$ is the result of one application of (2)). The sifting stops when the normalized squared difference between two successive iterates,

$$\text{SD}_k = \frac{\sum_t (h^{(k)}(t) - h^{(k-1)}(t))^2}{\sum_t (h^{(k-1)}(t))^2}, \quad (4)$$

falls below a prescribed threshold; Huang et al. [1] recommend $\text{SD} \in [0.2, 0.3]$, and note that this is a *very rigorous* limit, comparable to the change in a Fourier spectrum produced by shifting the data by only a few samples.

S-number criterion [1,9]. The sifting stops when, for S consecutive iterations, (i) the number of zero crossings equals the number of extrema or differs by at most one, and (ii) this count is *stable*. Extensive testing suggests $S \in [3, 8]$, with the lower end preferred. The optimal S can also be read off from a *confidence-limit* study: decompose the same data with different S and different stopping rules, and take the spread of the resulting IMFs as the uncertainty; the value of S at which the spread stops decreasing is the “optimal” one [9].

Both criteria are, in the language of numerical analysis, **a posteriori** fixed-point/Cauchy tests for the nonlinear iteration (2). Neither has a rigorous a priori justification that ties the

threshold to the error in the extracted IMF, and both make the decomposition *sensitive to local perturbations and to the amount of data* [2,9]: changing a few samples at the beginning of the record, or changing the stopping threshold, can change the entire set of IMFs. This sensitivity is one of the motivations for the ensemble methods.

2.4 Boundary (end) effects

Because the envelopes (2) are splines *interpolating* the observed extrema, they are least reliable near the endpoints, where there is no extremum to anchor the spline on one side. The result is an *end effect*: the extracted IMFs are distorted near $t = 0$ and $t = T$, and the distortion propagates inward over a distance comparable to the longest local wavelength. A variety of ad hoc remedies exist (endpoint extension, mirroring, boundary conditions on the spline), but none is universally satisfactory, and the end effect is a persistent, unavoidable feature of the spline-envelope construction.

3. A functional-analytic reformulation

The previous section described EMD as a recipe. For an analyst, the useful object is the **operator** that the recipe implements, and the question is what can be said about it in a function space.

3.1 EMD as a nonlinear iterative operator

A first caveat on the *setting*: EMD is not naturally defined on $L^2[0, T]$, because local extrema and spline envelopes depend on a *pointwise representative* of the signal, and L^2 functions are defined only up to null sets. The natural domain is a space of continuous (indeed, for the splines, C^1 or smoother) functions, e.g. $C([0, T])$ or $C^1([0, T])$ with its sup norm. We write the signal space as $X = C([0, T])$ for the nonlinear operator below; the L^2/ℓ^2 statements that follow apply to the *linear* idealization (6), which is well-defined on L^2 .

Define the **envelope operators**

$$E_+, E_- : X \rightarrow X,$$

mapping a signal to its upper and lower envelope (the cubic splines through its local maxima and minima), and the **mean-envelope operator**

$$M = \frac{1}{2}(E_+ + E_-).$$

The single sifting step (2) is the operator

$$T = I - M, \tag{5}$$

and one mode extraction is the repeated application $h^{(k+1)} = T h^{(k)}$ until the stopping rule. The full EMD is the composition of K such extractions, each applied to the residual left by the previous one.

The operators $E_{\pm}$ — and hence M and T — are **nonlinear** (they depend on the data's extremal structure) and are **not defined on all of X** (they require the presence of local extrema; a monotone or constant function has none). They are also **not contractive in any standard norm** that has been established; the sifting is not a projection, and T does not commute with itself in a way that would make the iterates a simple spectral sequence. This is precisely why EMD, unlike the Fourier or wavelet transform, has no general convergence theorem in the literature: the iteration (5) is a *data-dependent nonlinear* iteration,

and the tools that guarantee convergence of linear iterations (spectral radius < 1 , contractivity) do not directly apply.

3.2 The linear idealization: iterative filtering

The rigorous theory that *does* exist attaches to a **linear idealization** of the sifting [8,11,12,14]. The observation, made by Flandrin, Rilling, Gonçalves, and others, is that the mean-envelope subtraction (2) *behaves like a band-pass filter* whose frequency response is determined by the local spacing of the extrema: it passes the "fast" oscillations (spacings near the mean inter-extremum distance) and removes the slow trend. If one replaces the nonlinear M by a **linear** band-pass/low-pass filter P with frequency response $H(\omega) \in [0, 1]$ (where H is the fraction of the "fast" band removed per step), the sifting becomes the **linear iterative filtering** iteration

$$x^{(k+1)} = (I - P)x^{(k)} = (I - P)^k x. \quad (6)$$

This is the *Iterative Filtering (IF)* method, proposed around 2011 as a principled alternative to the spline EMD [11], and it is the object to which the convergence theorems apply.

3.3 Convergence of the linear idealization

The convergence of (6) is a clean result in L^2 (or ℓ^2). Decompose x into Fourier modes; on a single mode at frequency ω , the iteration (6) multiplies by $(1 - H(\omega))$ per step, so the k -th iterate of that mode is $(1 - H(\omega))^k x_\omega$, which tends to 0 **if and only if**

$$|1 - H(\omega)| < 1 \quad \Longleftrightarrow \quad 0 < H(\omega) < 2. \quad (7)$$

Because $0 \leq H \leq 1$ for a genuine band-pass, (7) holds automatically on the pass-band, and the convergence is **geometric** with rate $|1 - H(\omega)|$:

$$\|(I - P)^k x\|_{L^2} \leq \left(\sup_{\omega} |1 - H(\omega)| \right)^k \|x\|_{L^2}. \quad (8)$$

The **first extracted IMF** is the component of x in the band where $H \neq 0$ (the frequencies that are progressively removed); the residual retains the band where $H = 0$. In the notation of the recent two-frequency analysis [11], the theorem states that, given a filter w with normalized response $\widehat{w}$ and an aperiodic signal, if $|1 - \widehat{w}(\xi)| < 1$ (or $\widehat{w}(\xi) = 0$), then the iterative filtering converges and the first IMF is exactly the band-pass component of the signal. The condition (7) is the *a priori* guarantee: it is a condition on the *filter*, not on the data, and it is verifiable in advance.

This is the rigorous core of the whole story, and it is worth stating its scope honestly: **the convergence theorem is for the linear filter (6), not for the nonlinear spline sifting (5)**. The spline EMD is a *data-dependent approximation* of (6) whose effective $H(\omega)$ changes with the signal and with the iteration. The theorem tells us that *if* the spline sifting can be bounded by, or approximated by, a linear filter satisfying (7), then it inherits convergence; but bounding the nonlinear operator by a linear one, uniformly over the data, is not established in general.

3.4 The dyadic (octave-band) character

A further and important result of Flandrin, Rilling, and Gonçalves [8] is that the EMD sifting behaves, on average, as a **dyadic** (octave-band) filter: each IMF occupies a frequency band that is roughly one octave wide, and successive modes occupy successively lower

octaves. This was established by decomposing a wideband *fractional Gaussian noise* (fGn) model with spectrum

$$S(f) \propto |f|^{1-2H}, \quad 0 < H < 1, \quad (9)$$

where H is the Hurst exponent: the EMD of the fGn produces a cascade of octave-band modes, and the modes are “significant” (i.e., carry signal rather than just the $1/f$ -like noise) precisely where the empirical, mode-by-mode spectrum deviates from the fGn prediction (9). The upshot is that EMD is a **log-frequency multiresolution analysis** — structurally close to a wavelet or to a constant- Q (log-frequency) short-time Fourier analysis — but *nonlinear and data-adaptive*, with the octave boundaries determined by the data rather than fixed in advance. This is the sense in which EMD is “the natural basis” for nonlinear, nonstationary oscillations: it adapts the scale (wavelength) to the local content.

This dyadic property is not merely descriptive; it is *load-bearing* for CEEMDAN (Section 7.1), whose adaptive-noise construction relies on the k -th mode of white noise occupying the same octave band as the k -th mode of the signal.

3.5 What is, and is not, proven

To summarize the state of the theory for the true (spline) EMD:

- **Proven:** convergence of the *linear* iterative-filtering idealization (6) under the condition (7) [8,11,12,14]; the dyadic/octave-band character of the sifting on average [8]; a non-smooth convex-optimization reformulation of a related sifting with convergence guarantees [12].
- **Not proven (in general):** convergence of the fully nonlinear spline sifting (5); existence and uniqueness of the IMF set; a rigorous error bound tying the stopping threshold (4) to the error in the extracted mode; a separation theorem quantifying *when* EMD can resolve two close-by frequencies (the “one or two frequencies” question, addressed only recently and only for the linear IF method [11]).

The gap between “proven for the linear idealization” and “not proven for the true nonlinear operator” is the central theoretical limitation of EMD, and it is the gap that the ensemble methods are, in effect, a workaround for.

4. Pathologies: mode mixing, non-uniqueness, and the ad-hoc stopping rule

4.1 Mode mixing

Mode mixing [2,9] is the central pathology. It occurs in two related forms:

- *One mode, two frequencies:* a single IMF contains oscillations at two distinct instantaneous frequencies, because the sifting cannot decide which is “fastest” where the two overlap.
- *One frequency, two modes:* a single physical frequency is split across two or more IMFs.

In **running example (1)**, mode mixing is *very likely* in the overlap region I_0 , where $\varphi'_1(t) \approx \varphi'_2(t)$. There, the two tones are locally hard to distinguish on the inter-extremum scale, so the sifting (2) cannot cleanly separate them: the first IMF C_1 tends to “wobble” between tracking φ_1 and tracking φ_2 , and the envelope mean is contaminated by the slower tone. The result is an instantaneous frequency $\omega_{C_1}(t)$ that jumps between the two values over I_0 , and a mode C_1 that is not a clean single oscillation.

A word on the strength of this claim: **no theorem guarantees** that mode mixing must occur for two close-by frequencies. What the evidence and the literature establish is that the *risk* of mode mixing increases with (i) frequency proximity, (ii) intermittency of one of the

components, and (iii) amplitude disparity between the components; many close-frequency examples are successfully separated, particularly when the separation is maintained over a sufficient number of cycles. The statement “mode mixing is guaranteed” would be an overstatement; “very likely under proximity, intermittency, or amplitude disparity” is the defensible form. Outside I_0 , where the tones are separated, the sifting behaves as intended.

Mode mixing is not a bug that can be patched by a better spline or a better stopping rule; it is a *consequence of the nonlinearity and data-adaptivity*. The sifting has no *a priori* notion of a frequency band to enforce; it only knows the local extrema, and where two tones overlap, the local extrema carry the mixture.

4.2 Non-uniqueness and sensitivity

Because the stopping rule (4) or the S -number is a threshold, and because the sifting is nonlinear, the decomposition is **not unique**: different thresholds, different spline endpoint treatments, or a few extra/missing samples produce different IMF sets [2,9]. There is no canonical “the” EMD of a signal; there is a *family* of decompositions parameterized by the stopping rule and the endpoint treatment. This non-uniqueness is what the confidence-limit study [9] measures (the spread over the family) and what the ensemble methods aim to *average out*.

5. Ensemble EMD (EEMD)

5.1 The noise-assisted ensemble

Wu and Huang's **Ensemble Empirical Mode Decomposition (EEMD)** [2] regularizes the nonlinear operator by *stochastic perturbation*. The algorithm is:

1. Choose an ensemble size N and a noise amplitude σ (typically a fixed fraction, e.g. 0.1, of the standard deviation of x).
2. Draw N independent white-noise realizations $w^i(t)$, $i = 1, \dots, N$, each with standard deviation σ .
3. For each i , compute the **full** EMD of the noisy signal $x + w^i$, obtaining its complete set of IMFs $\{\text{IMF}_k^i\}_{k=1}^{K_i}$.
4. The k -th mode of the *original* signal is the **ensemble mean**

$$C_k = \frac{1}{N} \sum_{i=1}^N \text{IMF}_k^i. \quad (10)$$

The white noise serves two purposes. First, it *decorrelates* the mode mixing: adding independent noise to each copy perturbs the local extrema differently in each member, so the mixing “wobble” in C_1 is *independent* across members and cancels in the ensemble mean (10). Second, it makes the decomposition *robust*: the mean over many independent perturbations is far less sensitive to the stopping rule and to endpoint effects than any single decomposition.

In **running example (1)**, the effect of EEMD is that, in the overlap region I_0 , the per-member C_1^i still wobbles between φ_1 and φ_2 , but the *mean* C_1 is a stable, well-defined oscillation (roughly the energy-weighted combination of the two tones over I_0), and the instantaneous frequency $\omega_{C_1}(t)$ is smooth rather than jumping. The mode mixing is not *eliminated* (the two tones are genuinely not separable in I_0 by any method that only sees the data), but it is *averaged out* into a consistent, interpretable mode.

A subtlety that EEMD does not resolve, and that CEEMD and CEEMDAN will, is that **different members may yield a different number of modes** K_i [4]. The ensemble mean (10) is then ill-defined unless one pads the shorter decompositions with zeros, which

injects a residual noise and breaks the exact reconstruction. This is the “incompleteness” of EEMD, quantified in Section 5.2 and cured in Sections 6–7.

5.2 What EEMD estimates, and the $1/\sqrt{N}$ error reduction

A conceptual point deserves emphasis before the error analysis: EEMD does **not** estimate a “true” or “latent” IMF. There is no mathematically defined canonical decomposition of a nonlinear, nonstationary signal, so there is no latent object $\overline{C}_k$ toward which the ensemble converges in the way a Monte-Carlo integral converges to a fixed number. What EEMD estimates is the **ensemble mean of the (nonlinear) EMD operator applied to the noisy copies**,

$$C_k = \frac{1}{N} \sum_{i=1}^N \text{EMD}_k(x + w^i) \approx \mathbb{E}_w[\text{EMD}_k(x + w)],$$

i.e. the *average behavior* of the EMD operator under white-noise perturbation. This is a well-defined quantity (an expectation over the noise), and it is what the ensemble converges to; calling it the “true mode” would be a category error. The practical payoff is the same — a stable, interpretable, less mode-mixed oscillation — but the target is the noise-averaged operator output, not a hidden decomposition.

The error analysis is then a standard **Monte-Carlo / central-limit** argument [2]. Writing the sample mean as its expectation plus a fluctuation,

$$C_k = \mathbb{E}_w[\text{EMD}_k(x + w)] + \frac{1}{N} \sum_{i=1}^N \left(\text{EMD}_k(x + w^i) - \mathbb{E}_w[\text{EMD}_k(x + w)] \right),$$

and assuming the per-member deviations are approximately independent with variance σ_{IMF}^2 (a reasonable assumption for independent white-noise perturbations, though not a theorem for the nonlinear operator), the standard deviation of the error scales as

$$\text{std}(C_k - \mathbb{E}_w[\text{EMD}_k(x + w)]) \propto \frac{1}{\sqrt{N}}. \quad (11)$$

This is the standard Monte-Carlo scaling: the ensemble mean converges to the noise-averaged mode at rate $1/\sqrt{N}$. Wu and Huang [2] establish precisely this: for a fixed noise amplitude, increasing the ensemble size N reduces the standard deviation of the error in the ensemble-mean modes, and they give guidance that the noise amplitude σ should be chosen relative to the signal and that N should be large (hundreds to thousands) for a reliable decomposition. The scaling (11) is the *asymptotic* (law-of-large-numbers) result; a sharp, non-asymptotic bound for the *nonlinear* operator is not available, and the “independence” of the per-member deviations is an assumption, not a theorem (the sifting is nonlinear, so the per-member modes are not strictly independent of the noise in a way that a CLT directly covers). This is the honest limit of the EEMD error theory.

A concrete consequence, made explicit in [4], is the **residual-noise floor** of an *incomplete* EEMD: because the noise w^i is not explicitly removed and the mode count varies, the ensemble-mean reconstruction carries a residual noise of standard deviation

$$\varepsilon_r = \frac{\sigma}{\sqrt{N}}, \quad (11a)$$

so that recovering a reconstruction error below a target δ would require $N \gtrsim (\sigma/\delta)^2$ realizations. For $\sigma = 0.2$ and a machine-precision target $\delta \sim 10^{-15}$, this is $N \gtrsim 10^{29}$ — computationally absurd. This is the quantitative sense in which EEMD is “incomplete,”

and it motivates CEEMD's paired noise (Section 6) and CEEMDAN's exact reconstruction (Section 7).

5.3 Dyadic filter-bank interpretation

EEMD has a clean interpretation as a **randomized dyadic filter bank** [2,8]. Each ensemble member is a dyadic (octave-band) decomposition — by Section 3.4 — but the octave boundaries are *shifted* by the independent noise, so that each member samples a slightly different dyadic partition. The ensemble mean (10) is then a *coherent sum* over an ensemble of dyadic partitions, which averages out the partition-dependent (i.e., mixing-dependent) part and retains the common (i.e., signal) part. This is the filter-bank analogue of (11): the noise randomizes the filter boundaries, and the ensemble average is a variance-reduced estimate of the noise-averaged band-pass components.

5.4 Cost and parameter choice

The computational cost of EEMD is N **full** EMD decompositions (each with K_i modes and S sifting iterations per mode), i.e. of order $N \cdot K_i \cdot S$ envelope operations. This is the main practical limitation: a reliable EEMD often needs N in the hundreds to thousands, so the cost is N times that of a single EMD. The two free parameters are σ (noise amplitude) and N (ensemble size): too small a σ does not decorrelate the mixing, too large a σ injects noise that the ensemble cannot fully average; too small an N leaves residual variance (11). There is no rigorous, data-driven choice of (σ, N) ; in practice they are set by a target signal-to-noise ratio and a target error tolerance.

6. CEEMD: Complementary Ensemble EMD

6.1 The paired-noise construction

Complementary Ensemble Empirical Mode Decomposition (CEEMD), introduced by Yeh, Shieh, and Huang [3], is the intermediate step between EEMD and CEEMDAN. Its goal is to remove the *residual noise* that EEMD leaves behind, at essentially no extra cost, by adding the noise in **opposite-sign pairs**. The algorithm is:

1. Draw $N/2$ independent white-noise realizations w^i , $i = 1, \dots, N/2$, each with standard deviation σ .
2. For each i , compute the full EMD of the *pair* of noisy signals $x + w^i$ and $x - w^i$.
3. For each mode index k , average the four corresponding modes (two from $x + w^i$, two from $x - w^i$) over i :

$$C_k = \frac{1}{2(N/2)} \sum_{i=1}^{N/2} \left(\text{IMF}_k^{i,+} + \text{IMF}_k^{i,-} \right), \quad (12)$$

where $\text{IMF}_k^{i,\pm}$ are the k -th modes of $x \pm w^i$.

The key observation is that, to first order in the noise, the EMD operator satisfies $\text{EMD}_k(x + w^i) + \text{EMD}_k(x - w^i) \approx 2 \text{EMD}_k(x)$: the *odd* (linear-in-noise) part of the perturbation cancels between the two members of a pair, while the *even* part (the genuine mode) is retained. Averaging over pairs therefore removes the leading-order noise contribution from the ensemble mean, so the residual-noise floor is reduced relative to plain EEMD. The cost is the same as EEMD (still N full decompositions, just arranged in pairs), so CEEMD is a *variance-reduced* EEMD at no computational premium.

In **running example (1)**, CEEMD's effect is the same as EEMD's on the mode structure (the pairing does not change the mode-mixing behavior), but the residual noise in the reconstructed signal is smaller, because the paired averaging cancels the leading noise term.

6.2 Limitations

CEEMD reduces the residual-noise floor but does **not** enforce exact reconstruction: the mode count can still vary across members, and the telescoping sum (3) is not guaranteed to hold. It also does not change the *adaptive* character of the noise (the noise amplitude is still fixed at σ). These two limitations — incomplete reconstruction and non-adaptive noise — are precisely what CEEMDAN addresses.

7. CEEMDAN and ICEEMDAN

7.1 The CEEMDAN adaptive-noise construction

Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN), introduced by Torres, Colominas, Schlotthauer, and Flandrin [4], is the central method of this family. It modifies EEMD/CEEMD in two ways, both of which are essential: (i) it constructs the IMFs *sequentially* from ensemble averages with a **single, shared residual**, enforcing **exact reconstruction**; and (ii) it injects **adaptive** (octave-matched) noise at each stage. The algorithm, as given in [4], is:

Input: signal $x[n]$, ensemble size N , noise coefficients ε_k (chosen to fix the SNR at each stage).

1. **First mode.** Decompose by EMD the N realizations $x[n] + \varepsilon_0 w^i[n]$, $i = 1, \dots, N$, where w^i is white noise with $\mathcal{N}(0, 1)$; obtain their first modes and set

$$\widetilde{\text{IMF}}_1[n] = \frac{1}{N} \sum_{i=1}^N E_1(x[n] + \varepsilon_0 w^i[n]), \quad \tilde{r}_1[n] = x[n] - \widetilde{\text{IMF}}_1[n]. \quad (13)$$

Here $E_1(\cdot)$ denotes the operator that extracts the *first* EMD mode.

2. **Second mode.** Decompose the N realizations $\tilde{r}_1[n] + \varepsilon_1 E_1(w^i[n])$ until their first EMD mode, and set

$$\widetilde{\text{IMF}}_2[n] = \frac{1}{N} \sum_{i=1}^N E_1(\tilde{r}_1[n] + \varepsilon_1 E_1(w^i[n])). \quad (14a)$$

3. **k -th mode** ($k = 2, \dots, K$). Define the k -th residual

$$\tilde{r}_k[n] = \tilde{r}_{k-1}[n] - \widetilde{\text{IMF}}_k[n], \quad (14b)$$

then decompose the N realizations $\tilde{r}_k[n] + \varepsilon_k E_k(w^i[n])$ until their first EMD mode, and set

$$\widetilde{\text{IMF}}_{k+1}[n] = \frac{1}{N} \sum_{i=1}^N E_1(\tilde{r}_k[n] + \varepsilon_k E_k(w^i[n])). \quad (14)$$

4. **Stop** when the residual $\tilde{r}_K$ has fewer than two extrema (no further IMF can be extracted).

Output: the modes $\widetilde{\text{IMF}}_1, \dots, \widetilde{\text{IMF}}_K$ and the final residual

$$R[n] = x[n] - \sum_{k=1}^K \widetilde{\text{IMF}}_k[n], \quad x[n] = \sum_{k=1}^K \widetilde{\text{IMF}}_k[n] + R[n]. \quad (15)$$

The **adaptive-noise term** $\varepsilon_k E_k(w^i[n])$ in (14) is the “adaptive” in CEEMDAN: the noise added at stage k is *not* raw white noise of a fixed amplitude (as in EEMD), but the **k -th EMD mode of the white-noise realization w^i** , scaled by ε_k . This rests on the

dyadic/octave-band property of the EMD filter bank [8,13]: since the k -th EMD mode of *any* signal (white noise included) is concentrated in the k -th octave, the k -th mode of the white noise is, to the accuracy of the dyadic property, frequency-matched to the k -th mode of the signal, so the noise is matched to the octave band being extracted at each stage. The coefficients ε_k allow one to select the SNR at each stage independently; in [4] a fixed SNR is used across all stages.

Note on terminology. The algorithm above is the one given in the original ICASSP 2011 paper [4]. A closely related variant, **ICEEMDAN** (Section 7.4), replaces the fixed coefficient ε_k with a noise amplitude scaled to the *standard deviation of the first-mode ensemble*. Both share the “complete” reconstruction property (15) and the adaptive-noise idea; they differ only in the precise scaling of the injected noise.

7.2 Why “complete”: exact reconstruction

The defining property of CEEMDAN, and the reason for the “C,” is **exact reconstruction** (15). In EEMD and CEEMD, each member $x \pm w^i$ is decomposed *independently* in full, the noise is *discarded*, and — critically — different realizations may produce a *different number of modes* K_i , so the ensemble mean must be padded with zeros for the missing modes, introducing a residual noise of standard deviation $\varepsilon_r = \sigma/\sqrt{N}$ [4]. The reconstruction is therefore *inexact*: the sum of the ensemble-mean modes does not return x exactly, and the residual-noise floor (11a) sets a hard limit on the achievable reconstruction error for a given N .

In CEEMDAN, by contrast, a **single, consistent residual** $\tilde{r}_k$ is carried through all stages, and the telescoping sum (15) holds *exactly* by construction: each mode is subtracted from the same residual, so no noise is left behind and no padding is needed. The noise is *retained* in the residual and *re-injected* at each level in the frequency-matched form $\varepsilon_k E_k(w^i)$, so the noise budget is consistent across all levels. This internal consistency is what “complete” denotes. The reconstruction error in [4] is verified numerically to be at machine precision (max amplitude $< 2 \times 10^{-15}$, standard deviation 2×10^{-14} for an ECG signal), whereas matching that precision with EEMD would require $N \gtrsim 10^{29}$ realizations by (11a) — an impossibility.

It is worth being precise about the *priority* of the two CEEMDAN innovations. Exact reconstruction is the structural, load-bearing property: it is what makes the decomposition *invertible* and what removes the residual-noise floor that plagues EEMD and CEEMD. The adaptive (octave-matched) noise is a *refinement* that improves the *mode alignment* and *spectral separation* at each stage, on top of the exact reconstruction. Both matter, but the completeness is the primary innovation; the adaptivity is what makes CEEMDAN a *better* complete decomposition, not merely a complete one.

In **running example (1)**, CEEMDAN’s advantage over EEMD appears at the *deeper* modes. In the overlap region I_0 , the first mode $\widetilde{\text{IMF}}_1$ is the same stable, averaged oscillation in both EEMD and CEEMDAN. But when extracting $\widetilde{\text{IMF}}_2$ (the slower tone), EEMD decomposes the *original* noisy signal in full, so the noise in $\widetilde{\text{IMF}}_2$ is still at the original amplitude σ — mismatched to the smaller energy of the second residual — whereas CEEMDAN injects $\varepsilon_1 E_1(w^i)$, whose octave band is matched to the *second* mode by the dyadic property. The result is that CEEMDAN’s $\widetilde{\text{IMF}}_2$ is a cleaner estimate of the slower tone φ_2 , with less residual-noise contamination, and the mode mixing in the overlap region is more effectively suppressed.

7.3 CEEMDAN vs. EEMD/CEEMD

The distinctions can be summarized as follows:

Aspect	EEMD [2]	CEEMD [3]	CEEMDAN [4]
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Noise injection	fresh white noise, fixed σ	opposite-sign pairs, fixed σ	k -th EMD mode of the noise, $E_k(w^i)$, scaled by ε_k
Decomposition per member	full EMD, indepen- dently	full EMD, in pairs	first IMF only , per stage, on a shared residual
Residual	one per member (in- consistent)	one per member (in- consistent)	single, shared (con- sistent)
Reconstruction	inexact, floor (11a)	inexact, reduced floor	exact (15)
Noise cancellation	by ensemble aver- aging	by ensemble aver- aging + paired can- cellation	by ensemble av- eraging + octave- matched noise
Mode alignment	baseline	same as EEMD	improved (frequency- matched noise)
Cost	$N \cdot K_i \cdot S$	$N \cdot K_i \cdot S$	$K \cdot N \cdot S$
Effective N needed	larger	smaller (noise floor)	smaller (complete- ness + alignment)

The computational orders are comparable (all are N times a per-level cost), but CEEMDAN typically achieves a given quality with a *smaller* N , because the adaptive noise matching makes each ensemble member a better estimate. In [4], for a 512-sample Dirac signal, CEEMDAN required about half the sifting iterations of EEMD at $N = 500$, and for a 10-second ECG signal only 30.8% of the EEMD iterations. The practical consequence is that CEEMDAN is often the preferred method when the signal has well-separated scales and one wants a clean, internally-consistent, exactly-reconstructing decomposition with less mode mixing, at a cost comparable to (and often lower than) EEMD.

7.4 ICEEMDAN: the improved variant

Improved Complete Ensemble EMD with Adaptive Noise (ICEEMDAN), introduced by Colominas, Schlotthauer, and Torres [5], is the current workhorse of the family in much of the applied literature. It refines CEEMDAN in one specific way: instead of using a fixed noise coefficient ε_k , the noise injected at each stage is scaled to the **standard deviation of the first-mode ensemble** obtained in that stage. Concretely, after computing the first-mode ensemble $\{\text{IMF}_k^i\}_{i=1}^N$ at stage k , one computes its standard deviation σ_k (across the ensemble, at each time point) and injects noise of amplitude σ_k into the residual when extracting the next mode:

$$\widetilde{\text{IMF}}_{k+1}[n] = \frac{1}{N} \sum_{i=1}^N E_1(\tilde{r}_k[n] + \sigma_k w^i[n]), \quad \sigma_k[n] = \text{std}_i\{\text{IMF}_k^i[n]\}. \quad (16)$$

The rationale is that σ_k measures the *actual* spread of the k -th mode ensemble — a data-driven estimate of how much noise is needed to perturb the sifting at that scale — rather than a fixed, user-chosen coefficient. This makes the noise *adaptive to the data* in a stronger sense than CEEMDAN: the amplitude tracks the ensemble's own variability, so stages with a well-determined mode (small σ_k) get less noise, and stages with a noisy, uncertain mode (large σ_k) get more. The practical benefits, reported in [5] and subsequent work, are: (i) **reduced residual noise** in the modes, (ii) **improved mode separation** (less mode mixing at the boundaries), (iii) **reduced spurious modes** (the data-driven scaling avoids over-perturbing stages where the mode is already well determined), and (iv) **improved reconstruction accuracy** (still exact, but with less noise contamination in the individual modes).

ICEEMDAN is therefore the most refined member of the family for many practical applications, and it is the variant most commonly implemented in open-source toolboxes. The cost is comparable to CEEMDAN (the extra work is computing the per-stage ensemble standard deviation, which is negligible relative to the EMD decompositions).

8. A unified numerical-analysis viewpoint

The five methods are best understood as a single object — a nonlinear decomposition operator $\mathcal{E}$ — refined in a sequence, each with a precise numerical-analysis interpretation.

8.1 (Regularized) nonlinear operators on $C([0, T])$

- **EMD** is the raw nonlinear operator $\mathcal{E}$ (the composition of the sifting operator $T = I - M$ (5) over K stages). It is a data-dependent, non-projection, non-contractive operator on a *subset* of $X = C([0, T])$ (the functions with enough extrema). It has no general convergence theorem.
- **EEMD** is $\mathcal{E}$ **regularized by Monte-Carlo averaging**: one evaluates the (nonlinear) mode-extraction map at N independent noise-perturbed inputs and averages, estimating $\mathbb{E}_w[\mathcal{E}(x + w)]$. The error is controlled by the variance of the perturbed outputs (11).
- **CEEMD** is the same Monte-Carlo regularizer with **paired (antipodal) noise** $(+w^i, -w^i)$, so that the leading (linear-in-noise) error cancels in each pair — a variance-reduced EEMD at the same cost.
- **CEEMDAN** is the Monte-Carlo regularizer with **sequential construction** (single shared residual, exact reconstruction (15)) and **adaptive (octave-matched) noise** $E_k(w^i)$, scaled by ε_k (14). It is a *closed-loop* version of the open-loop EEMD regularizer: the noise at stage k depends on the residual left by stages $1, \dots, k - 1$.
- **ICEEMDAN** is CEEMDAN with the noise amplitude replaced by the *data-driven* per-stage ensemble standard deviation σ_k (16), making the adaptation fully data-driven.

In the language of numerical analysis, EMD is an *unregularized* nonlinear fixed-point iteration; EEMD is a *Monte-Carlo* (stochastic) approximation to its output; CEEMD is a *variance-reduced* Monte-Carlo approximation; CEEMDAN is a *stochastic* approximation with *exact telescoping* and *adaptive step-size*; ICEEMDAN is the same with a *data-driven step-size*. This is an apt analogy: the noise amplitude plays the role of a step-size that is scaled to the local residual (CEEMDAN) or to the ensemble's own variability (ICEEMDAN), exactly as in an adaptive iterative method.

8.2 Monte-Carlo variance reduction for a nonlinear operator

The error of EEMD, CEEMD, CEEMDAN, and ICEEMDAN is, to first order, a **variance-reduction** problem. The per-member modes are random variables (random in the noise), and the ensemble mean is their empirical mean. The standard Monte-Carlo result — variance $\propto 1/N$, standard deviation $\propto 1/\sqrt{N}$ — applies *conditionally* on the (approximate)

independence of the per-member deviations, which is an assumption for the nonlinear operator but a reasonable one for independent white-noise perturbations. The $1/\sqrt{N}$ scaling (11) is therefore the correct *order* of the convergence, even if a sharp non-asymptotic bound for the nonlinear operator is not available. CEEMD reduces the *constant* in (11) by cancelling the leading noise term; CEEMDAN and ICEEMDAN reduce it further by the octave-matched / data-driven noise scaling, so a smaller N suffices for a given accuracy.

8.3 The stopping criterion as a Cauchy/fixed-point test

The SD criterion (4) and the S -number are, in numerical-analysis terms, **a posteriori** Cauchy/fixed-point tests for the nonlinear iteration (5). They measure the change between successive iterates and stop when it is small — the standard *a posteriori* criterion for a fixed-point iteration. What they lack is an *a priori* bound tying the threshold to the *error in the extracted mode* (as opposed to the *change between iterates*), and this is the sense in which the stopping rule is ad hoc. The ensemble methods do not remove this problem — each member is still sifted with the same ad-hoc rule — but they *distribute* the sensitivity: because the stopping-rule error is (approximately) independent across members, it is averaged out in the ensemble mean (10)/(14), exactly as the mode-mixing error is.

8.4 Relation to wavelets and other multiresolution methods

The methods sit in the broader family of multiresolution analyses, and the contrast with the linear methods is instructive:

- **Wavelets / STFT:** linear, fixed, shift-invariant basis; exact convergence and stability (the basis is a frame/Riesz basis); the scale is *fixed in advance* (dyadic for wavelets).
- **EMD:** nonlinear, data-adaptive, no fixed basis; the scale is *adapted to the data* (dyadic on average, Section 3.4); no general convergence theorem; the mode-mixing and non-uniqueness are intrinsic.
- **EEMD / CEEMD / CEEMDAN / ICEEMDAN:** the nonlinear EMD operator *regularized* by stochastic perturbation; the convergence is the *Monte-Carlo* convergence of the regularized (averaged) operator, at rate $1/\sqrt{N}$; the price is a stochastic (not deterministic) error and an $O(N)$ cost.

The deep point for the analyst is that EMD and its ensemble refinements are the *nonlinear, data-adaptive* counterpart of the wavelet/STFT multiresolution analysis, and that the ensemble methods are the *stochastic regularization* that restores — at a Monte-Carlo rate and an $O(N)$ cost — much of the robustness and uniqueness that the linear methods enjoy by construction.

8.5 Summary of what is rigorous

Claim	Status
Convergence of the linear iterative-filtering idealization (6) under (7)	Proven [8,11,12,14]
Dyadic/octave-band character of the sifting (on average)	Proven (fGn analysis) [8]
EEMD/CEEMD/CEEMDAN/ICEEMDAN error $\propto 1/\sqrt{N}$	Proven in order (Monte-Carlo/CLT, conditional on approximate independence) [2]
Exact (machine-precision) reconstruction of CEEMDAN/ICEEMDAN	Proven by construction (telescoping (15)); verified numerically [4,5]
Convergence of the fully nonlinear spline sifting (5)	Not proven in general

Existence/uniqueness of the IMF set	Not proven in general
A priori bound tying the stopping threshold to the mode error	Not proven (ad-hoc, a posteriori rule)
Separation of two close-by frequencies by the true EMD	Not proven (only for the linear IF method, recently) [11]

9. Modern competitors: VMD and synchrosqueezing

For a numerical-analysis audience, it is worth situating the EMD family relative to the two most important modern competitors, both of which have a firmer mathematical footing than classical EMD.

9.1 Variational Mode Decomposition (VMD)

Variational Mode Decomposition (VMD), introduced by Dragomiretskiy and Zosso [6], is a *variational* (optimization-based) alternative to EMD. Rather than extracting modes adaptively by sifting, VMD formulates the decomposition as a **constrained variational problem**: given a signal $x(t)$ and a number of modes K , find K band-limited modes $u_k(t)$ with center frequencies ω_k and a residual $r(t)$ such that $\sum_{k=1}^K u_k(t) + r(t) = x(t)$ and the *sum of the bandwidths* of the modes is minimal. The bandwidth of each mode is estimated by the L^2 norm of the partial derivative of its demodulated (analytic) signal. The problem is solved by an **Alternating Direction Method of Multipliers (ADMM)** in the Fourier domain, with parameters: the number of modes K (user-chosen), a bandwidth penalty α , and a Lagrange multiplier relaxation ρ .

The key contrasts with EMD are:

Aspect	EMD	VMD
Formulation	nonlinear, adaptive, sifting	variational (convex sub-problems), ADMM
Number of modes	data-driven (adaptive)	user-chosen K
Mathematical footing	no general convergence theorem	ADMM convergence for the variational subproblems
Bandwidth control	implicit (dyadic, on average)	explicit (the α penalty)
Noise robustness	poor (mode mixing)	good (the variational formulation is noise-robust)
Adaptivity	fully data-driven	semi-adaptive (center frequencies are optimized, but K is fixed)

VMD is arguably the most important *competitor* to EMD for a numerical-analysis audience, precisely because it has a variational formulation and explicit bandwidth control. Its limitation is that the number of modes K must be specified in advance, which is a significant

loss of adaptivity relative to EMD's data-driven mode count. In practice, K is often chosen by a heuristic (e.g., minimizing a reconstruction error or a mode-variance criterion), which reintroduces an ad hoc element.

9.2 Synchrosqueezing

Synchrosqueezed transforms (synchrosqueezed wavelet transform, synchrosqueezed STFT), introduced by Daubechies, Lu, and Wu [7], are *reassignment*-based time-frequency methods that, unlike EMD, rest on a **linear and shift-invariant** foundation. The idea is to take a standard time-frequency representation (a continuous wavelet transform or a STFT) and *reassign* each coefficient to a refined time-frequency location (its local "ridge" frequency), thereby concentrating the representation and recovering a sharper time-frequency image. The result is a set of "modes" — one per ridge — that are, in the monocomponent case, close to the EMD IMFs, which is why Daubechies et al. describe synchrosqueezing as "an empirical mode decomposition-like tool."

A precision on linearity is in order. The underlying transform (STFT or CWT) is strictly linear, but the *reassignment rule itself* — the squeezing of coefficients $(t, \omega) \mapsto (t, \omega_{\text{ridge}})$ — is a **nonlinear** operation on the time-frequency plane: the destination of each coefficient depends on the local content of the representation. This is exactly why synchrosqueezing can sharpen ridges, and also why it can introduce **cross-term artifacts** when the components are not separated by the ϵ -bounds required by the convergence proofs of Daubechies et al. [7]. Synchrosqueezing is thus a *linear-based* method — linear at the foundation, nonlinear at the reassignment — in contrast to EMD's purely nonlinear foundation.

The key contrasts with EMD are:

Aspect	EMD	Synchrosqueezing
Linearity	nonlinear	linear foundation (reassignment rule is nonlinear)
Shift invariance	not shift-invariant	shift-invariant
Basis	data-driven, no fixed basis	fixed (wavelet or STFT) + reassignment
Convergence	no general theorem	guaranteed under the ϵ -separation bounds [7]
Adaptivity	fully data-driven	fixed basis (not adaptive to the signal)
Mode mixing	yes	no (no sifting; ridge artifacts only when separation bounds fail)

Synchrosqueezing is the *linear-based* counterpart of EMD: it achieves a similar multiresolution (mode-by-mode) decomposition on a linear, shift-invariant foundation, with convergence and stability guarantees — valid under the separation assumptions of [7] — at the price of losing the data-adaptivity. The recent two-frequency analysis [11] compares EMD, iterative filtering, and synchrosqueezing on the question of how well two close-by frequencies can be separated, and finds that the methods have complementary strengths: EMD/IF are more adaptive, synchrosqueezing is more stable.

9.3 The trade-off

The EMD family and its competitors embody a fundamental trade-off in multiresolution analysis: **adaptivity vs. rigor**. EMD and its ensemble refinements are fully data-adaptive (no fixed basis, data-driven mode count) but lack a general convergence theorem; VMD and synchrosqueezing are more rigorous (variational, or linear-based with convergence

under separation bounds) but sacrifice some adaptivity (VMD requires a fixed K ; synchrosqueezing uses a fixed basis). The ensemble methods (EEMD, CEEMD, CEEMDAN, ICEEMDAN) are the EMD family's answer to the rigor gap: they restore robustness and (approximate) uniqueness via stochastic regularization, at a Monte-Carlo rate and an $O(N)$ cost, but they do not remove the fundamental nonlinearity that is the source of the theoretical gap.

10. Current directions and open problems

The theory of EMD and its ensemble refinements is, as of the recent two-frequency analysis [11] and the convex-optimization reformulations [12], at the stage where the *linear idealization* is well understood but the *true nonlinear operator* is not. The principal open problems are:

1. **General convergence of the nonlinear sifting.** Establishing, for the spline operator $T = I - M$ (5), an *a priori* convergence and stability result that does not rely on bounding T by a linear filter. The convex-optimization reformulation of a related sifting [12] is a promising route, because it recasts the sifting as a (non-smooth) convex minimization, for which standard convergence theorems apply; extending this to the exact spline sifting is an open problem.
2. **The two-frequency separation limit.** Quantifying, for the *nonlinear* EMD (not just the linear IF), the condition under which two close-by frequencies are resolvable — the “one or two frequencies” question [11]. For the linear method this is governed by the filter response (7); for the nonlinear method the condition must involve the data-dependent envelope structure, and no such condition is known.
3. **A rigorous stopping criterion.** Replacing the ad-hoc SD/ S -number with an *a priori* rule whose threshold can be tied to the error in the extracted mode, and whose effect is *controllable* under the ensemble averaging of EEMD/CEEMD/CEEMDAN/ICEEMDAN.
4. **Non-asymptotic error bounds for the ensemble methods.** The $1/\sqrt{N}$ scaling (11) is an asymptotic (CLT) result conditional on approximate independence; a *non-asymptotic*, distribution-free bound for the nonlinear operator — e.g. via concentration inequalities that do not require independence — is an open problem that would make the choice of N principled rather than heuristic.
5. **Optimal noise scheduling in CEEMDAN/ICEEMDAN.** The choice of the adaptive-noise coefficients ε_k (CEEMDAN) or the data-driven scaling σ_k (ICEEMDAN) is a natural, heuristic choice; whether it is *optimal* in a variance-minimization sense, and whether a different schedule (e.g. one derived from a minimax bound, or one that matches the octave-band energy of the residual) can reduce the required N , is open. The original authors flag this explicitly as future work [4,5].

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A note on scope and honesty. Throughout this essay the distinction between what is **proven** (the convergence of the linear iterative-filtering idealization; the dyadic character; the $1/\sqrt{N}$ Monte-Carlo scaling; the exact reconstruction of CEEMDAN/ICEEMDAN by construction) and what is **heuristic or open** (the convergence of the fully nonlinear spline sifting; the existence/uniqueness of the IMF set; the optimality of the stopping rule and the CEEMDAN/ICEEMDAN noise schedule; the two-frequency separation limit for the true EMD; the Bedrosian-type assumptions underlying the instantaneous frequency) is maintained deliberately. The ensemble methods are, in effect, a *practical* workaround for the lack of a general nonlinear convergence theory: they restore robustness and (approximate) uniqueness at a Monte-Carlo rate and an $O(N)$ cost, but they do not remove the fundamental nonlinearity that is the source of the theoretical gap.